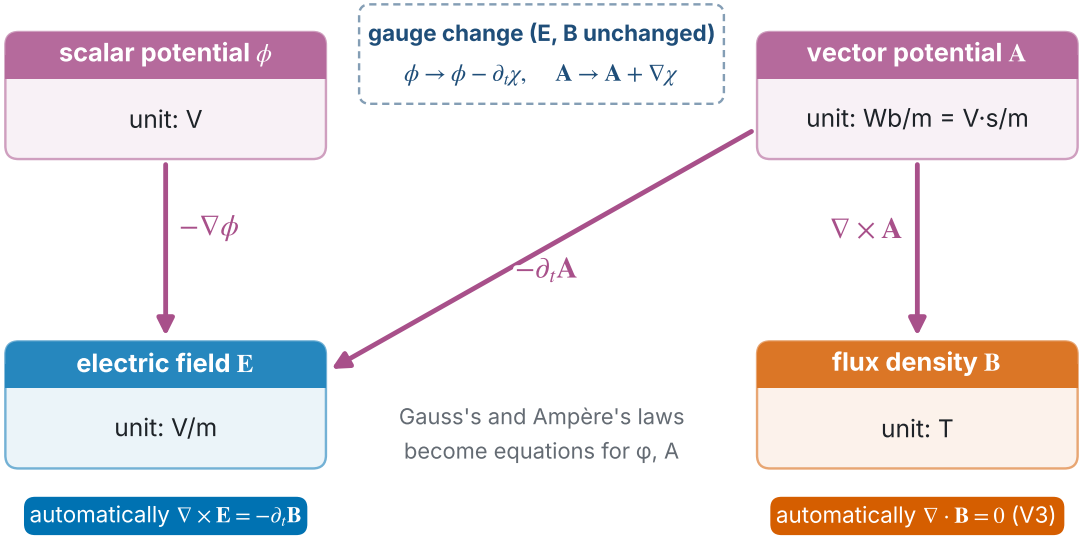
scalar potential ϕ
unit: V
gauge change (E, B unchanged)
$\phi \to \phi - \partial_t \chi, \quad \mathbf{A} \to \mathbf{A} + \nabla\chi$
vector potential $\mathbf{A}$
unit: Wb/m = V·s/m
$-\nabla\phi$
$-\partial_t \mathbf{A}$
$\nabla \times \mathbf{A}$
electric field $\mathbf{E}$
unit: V/m
Gauss's and Ampère's laws
become equations for φ, A
flux density $\mathbf{B}$
unit: T
automatically $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$
automatically $\nabla \cdot \mathbf{B} = 0$ (V3)